Power

- Power is the rate of which work is done.
- $P = W/\Delta t$
  - W: work in Joules
  - Δt: elapsed time in seconds
- When we run upstairs, t is small so P is big.
- When we walk upstairs, t is large so P is small.

Unit of Power

- SI unit for Power is the Watt.
  - 1 Watt = 1 Joule/s
  - Named after the Scottish engineer James Watt (1776-1819) who perfected the steam engine.
- British system
  - horsepower
  - 1 hp = 746 W

Sample problem

A man runs up the 1600 steps of the Empire State Building in 20 minutes. If the height gain of each step was 0.20 m, and the man's mass was 80.0 kg, what was his average power output during the climb? Give your answer in both watts and horsepower.

213.333 Watts

0.286 hp

Sample problem

Calculate the power output of a 0.10 g fly as it walks straight up a window pane at 2.0 cm/s.

2 E -5 Watts